Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Analysis of Talabat Delivery Orders

Submitted by: Haya Mohamed Moussa (24-101252)

Yassin Fouad (24-101177)

Zyad Walid(24-101301)

Omar Khattab (24-101227)

Introduction

In today's fast-paced world, food delivery platforms like Talabat have become an essential part of daily life across the Middle East, handling hundreds of thousands of orders every day in a market where customer satisfaction is everything. But behind every successful delivery is a complex system of decisions — which driver to assign, which route to take, and how conditions like traffic affect timing.

This project analyzes a dataset of Talabat delivery orders using exploratory data analysis, visualization, and statistical hypothesis testing to uncover hidden patterns in customer behavior and delivery performance. By answering these questions with data, we move beyond assumptions and into evidence-based insights that can drive real improvements in delivery efficiency and customer experience.

Research Question

What factors influence delivery performance and customer behavior in Talabat food delivery orders?

Hypothesis

Population of Interest:

The population of interest is all food delivery orders placed through the Talabat platform across all cities where it operates.

Sampling Method:

The sampling method is convenience sampling, as we used a readily available dataset obtained from the internet. It was the most practical one for this research topic since collecting real data from Talabat would require access to their internal systems, which is not something available to us as researchers.

Bias Identification:

Since this study used a pre-existing dataset rather than a survey designed by us, one of the main potential sources of bias is confirmation bias.

Confirmation bias occurs when researchers tend to focus on results that support their prior expectations while ignoring evidence that contradicts them. For example, we might assume that higher traffic levels always lead to longer delivery times and unintentionally interpret the data in a way that supports this belief. To minimize this bias, we followed a

structured hypothesis testing approach. Specifically, we defined null and alternative hypothesis before analyzing the data and used a fixed significance level of $\alpha = 0.05$. This approach ensures that decisions are based on statistical evidence (p-values) rather than personal expectations or assumptions.
A clear example of this is the traffic level analysis. Although we initially expected traffic to significantly impact delivery duration, the statistical test showed no significant difference between traffic levels. Instead of forcing our expectation onto the results, we accepted the outcome and failed to reject the null hypothesis.
This demonstrates that the analysis was driven by the data rather than personal beliefs, helping to reduce the impact of confirmation bias and improve the reliability of the conclusions.

Survey Questions/Collected Data/Dataset:

The dataset we used is a Talabat food delivery orders dataset obtained from the internet, containing **100,000 orders** and **23 columns** covering every aspect of the delivery process from the moment an order is placed to the moment it is delivered.

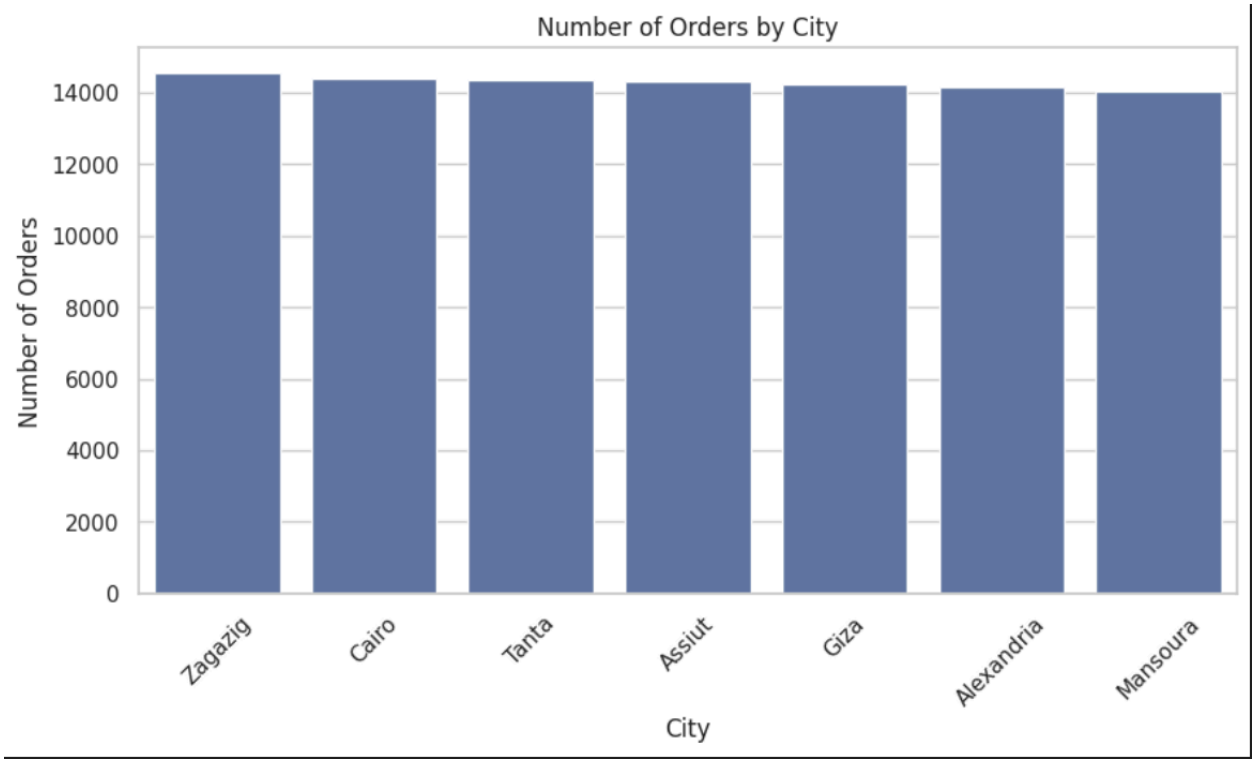
Features (Variables) Explanation:

- **Order_ID**: Unique identifier for each order
- **User_ID**: Unique identifier for each customer
- **Restaurant_ID**: Identifier for the restaurant
- **Driver_ID**: Identifier for the delivery driver
- **Item_Name**: Type of food ordered (e.g., Pizza, Burger, Sushi)
- **Quantity**: Number of items ordered
- **Total_Price**: Total cost of the order
- **Order_Time**: Timestamp when the order was placed
- **Delivery_Time**: Timestamp when the order was delivered
- **Delivery_Duration_Minutes**: Time taken for delivery
- **City**: City where the order was made
- **Payment_Method**: Method used (Cash, Wallet, Credit Card)
- **Order_Status**: Status of the order (Delivered, Cancelled, In Transit)
- **Driver_Vehicle**: Type of vehicle used (Car, Motorbike, Bicycle)
- **Restaurant_Lat / Restaurant_Lon**: Restaurant location
- **Customer_Lat / Customer_Lon**: Customer location
- **Driver_Lat / Driver_Lon**: Driver location
- **Delivery_Distance_km**: Distance between restaurant and customer
- **Traffic_Level**: Traffic conditions (Low, Medium, High)

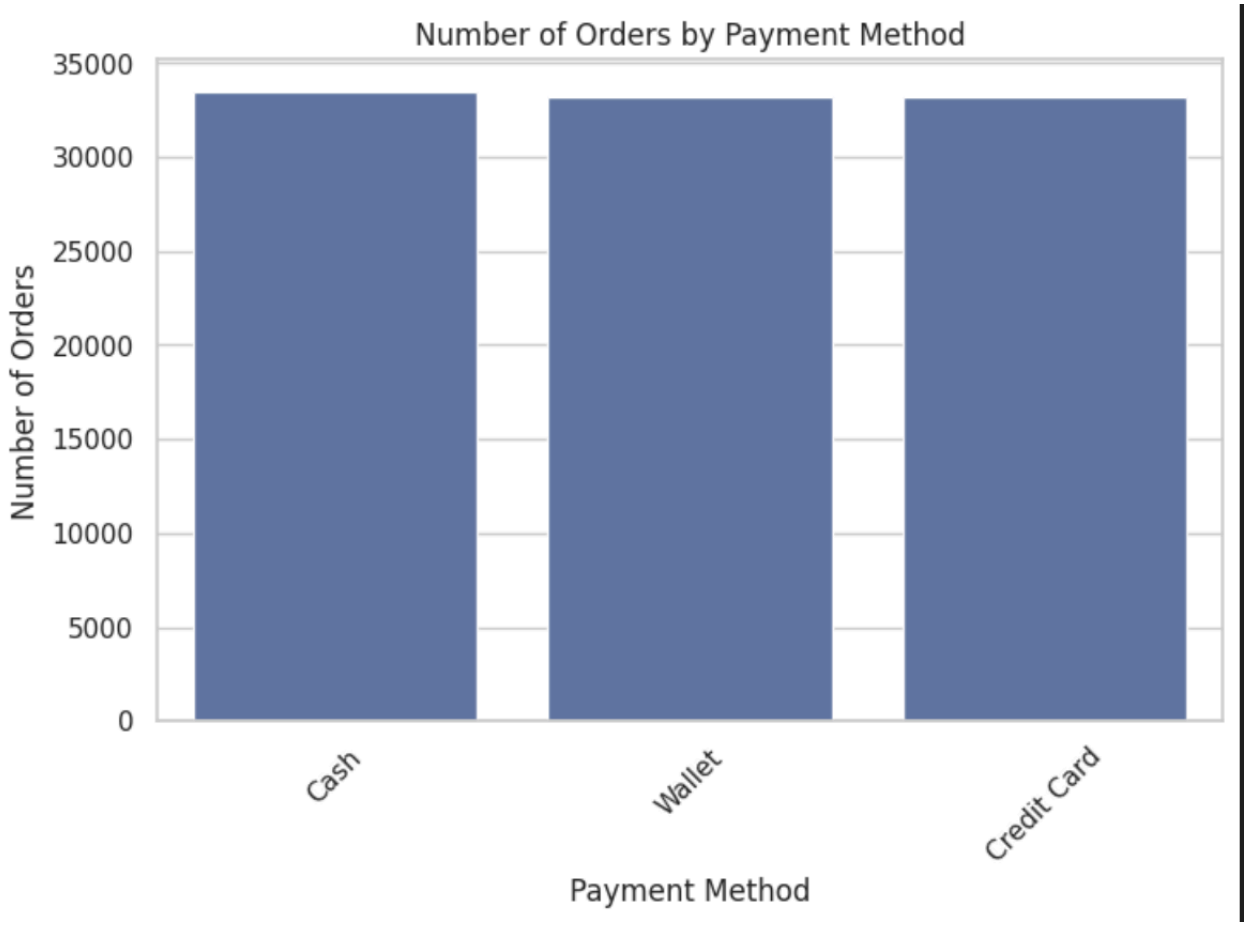
- **Driver_Availability**: Whether driver was Online or Offline

Numbers of samples used is: 10,000 samples

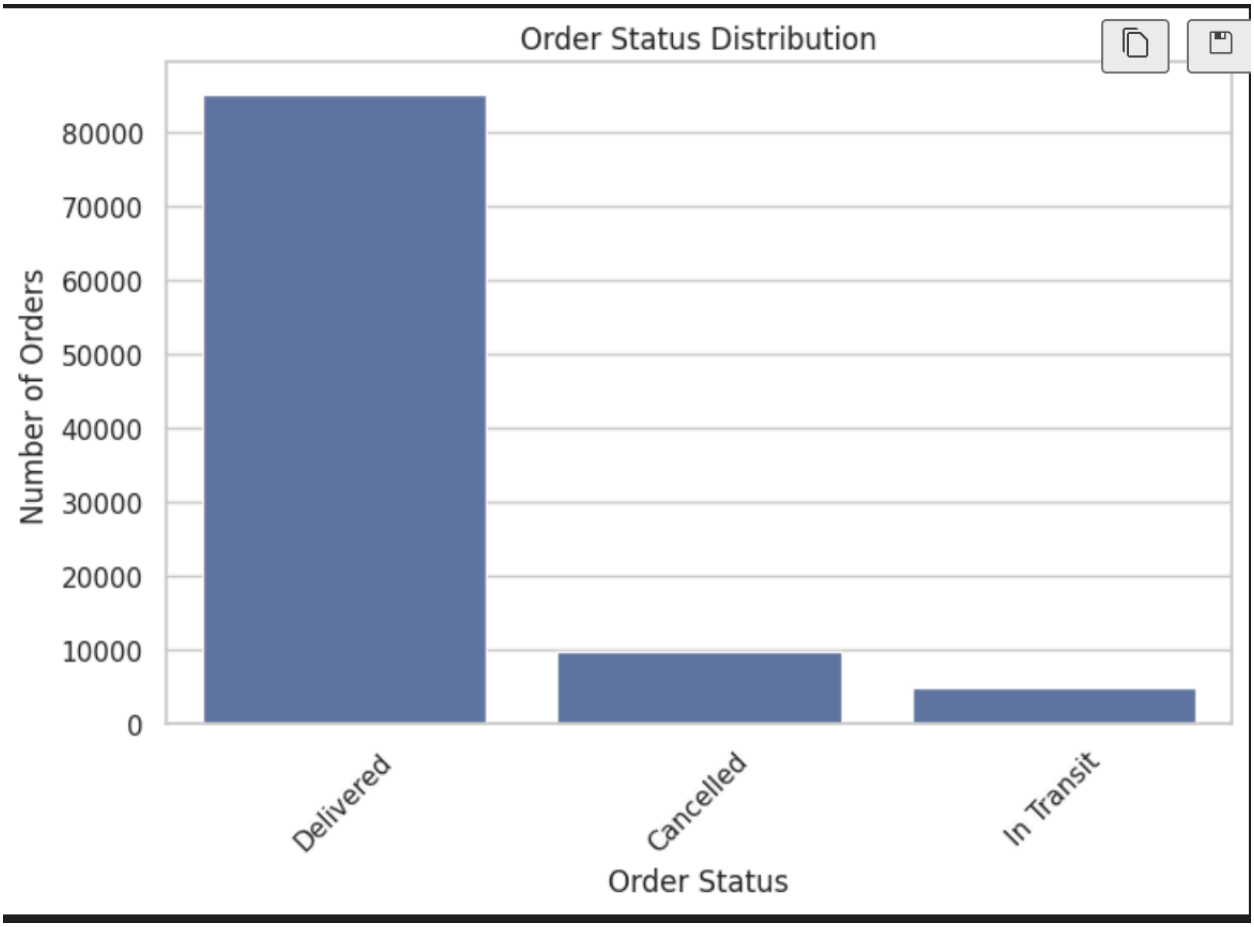
Analysis:



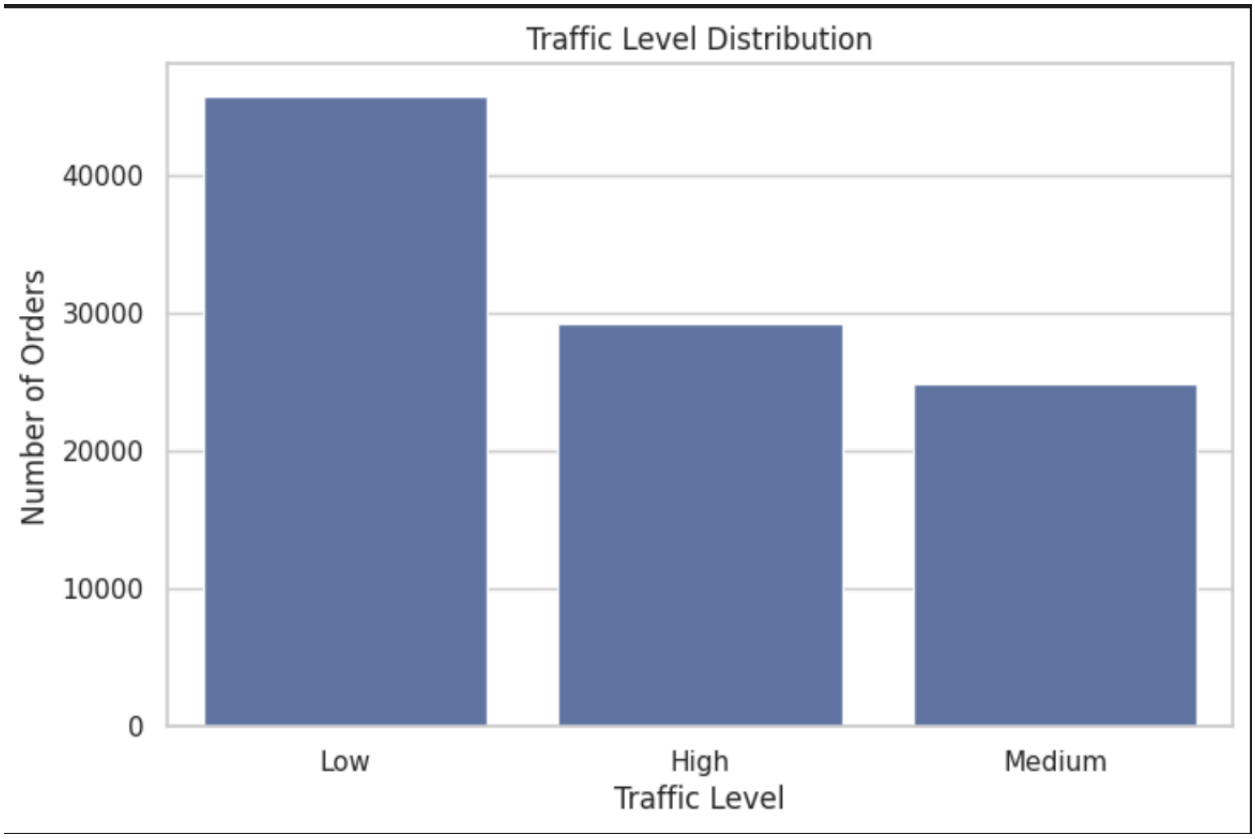
This count plot shows the number of orders across different cities. Orders are fairly evenly distributed across cities, meaning there is no strong geographical bias in demand. This improves the reliability of comparisons across cities.



This graph shows how frequently each payment method (Cash, Wallet, Credit Card) is used across all orders. The distribution is almost equal across all payment methods, indicating that customers do not strongly prefer one payment option over another. This suggests that all payment systems are equally accessible and widely adopted.

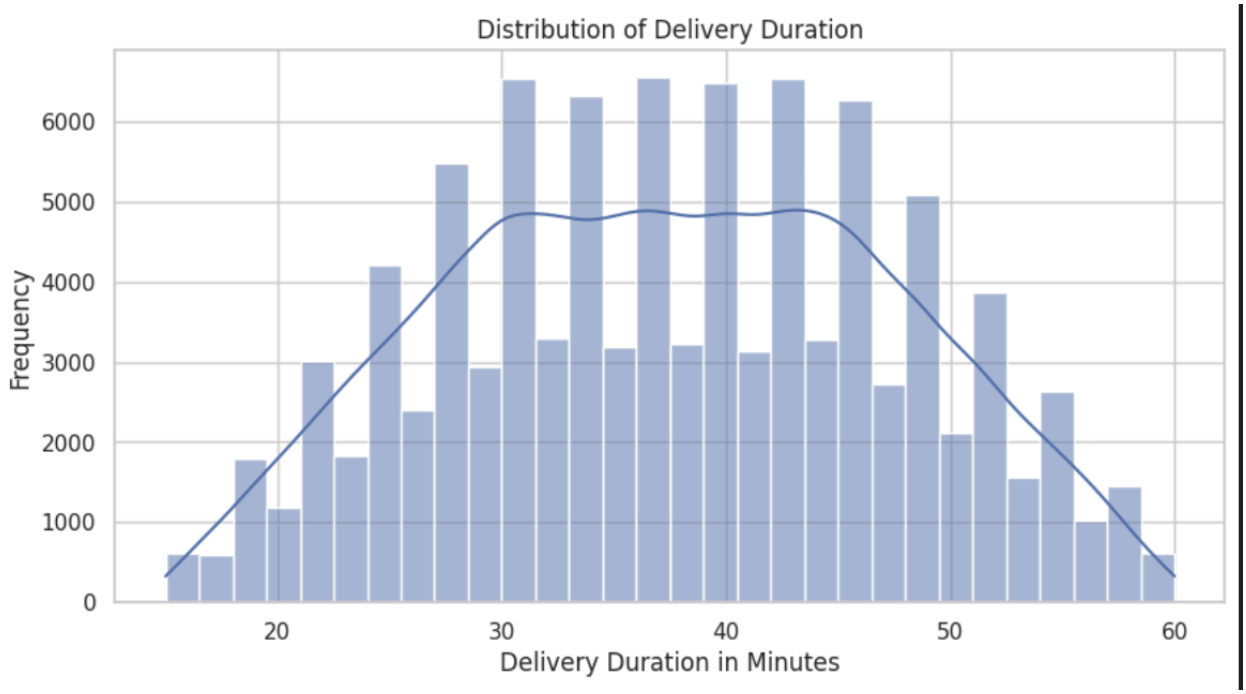


This chart displays the number of orders for each order status category: Delivered, Cancelled, and In Transit. Most orders are successfully delivered, with a smaller portion being cancelled or still in transit. This indicates that the delivery system is efficient and reliable with a high completion rate.

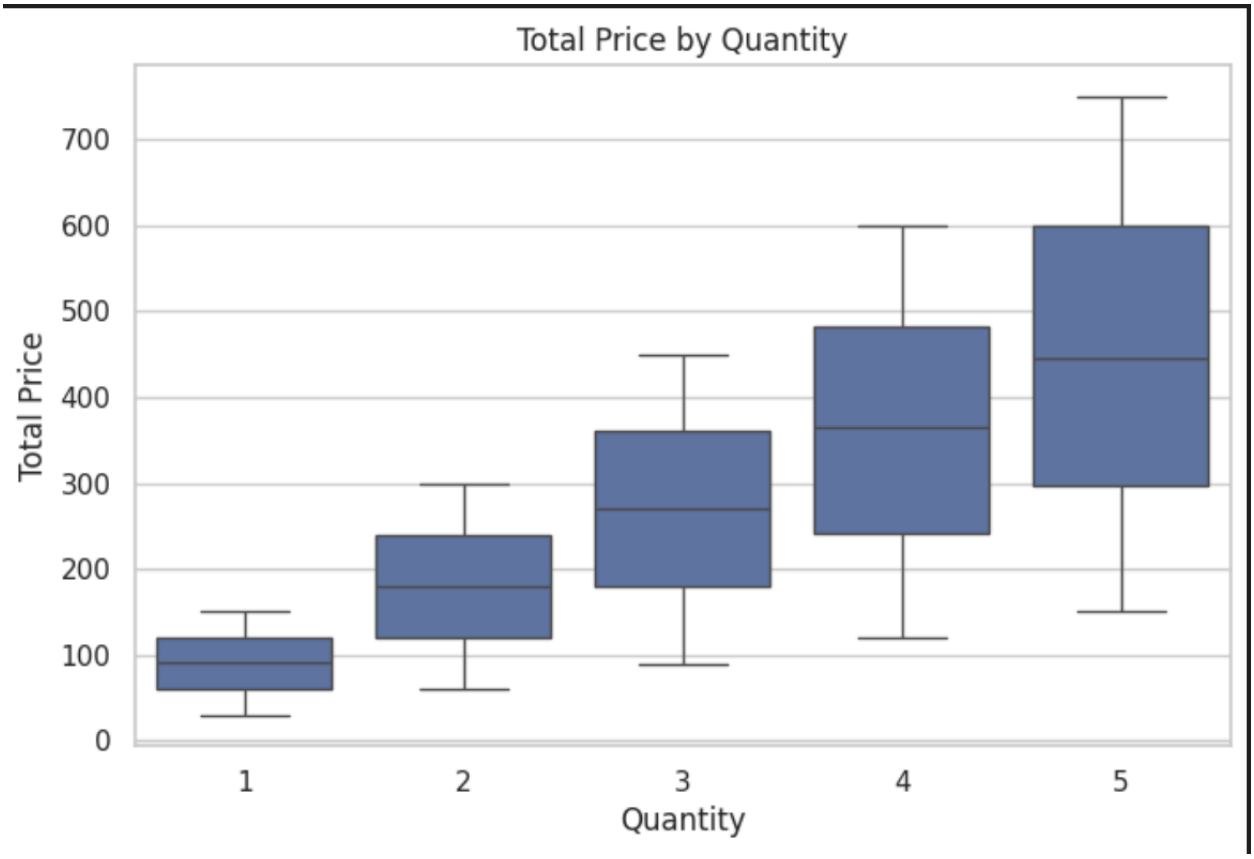


This graph shows how orders are distributed across different traffic conditions: Low, Medium, and High. Low traffic appears most frequently, while high traffic occurs less often. This suggests that most deliveries happen under relatively smooth traffic conditions, which may influence overall delivery performance.

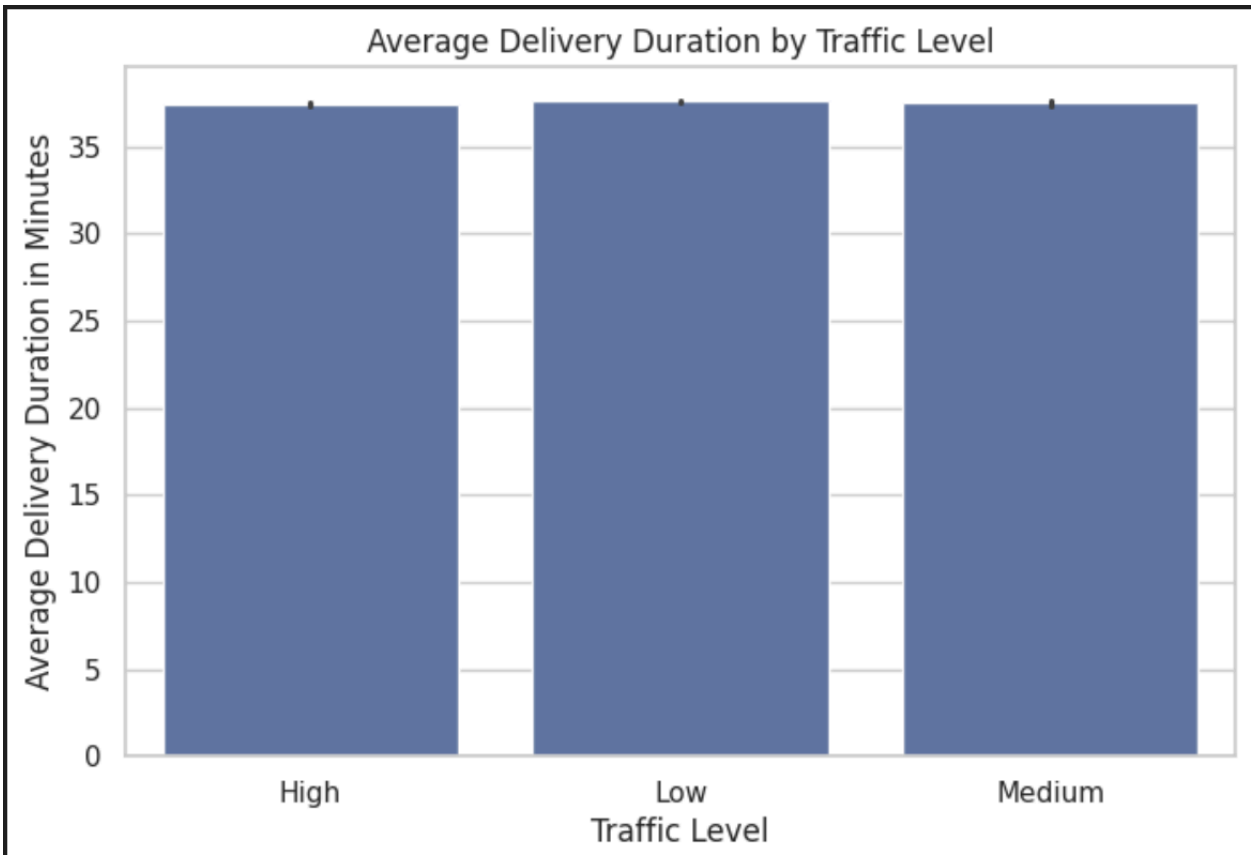
This histogram shows how total order prices are distributed across all transactions. Most orders fall within a moderate price range, with fewer extremely high or low values. This indicates that customer spending behavior is relatively consistent and not highly skewed.



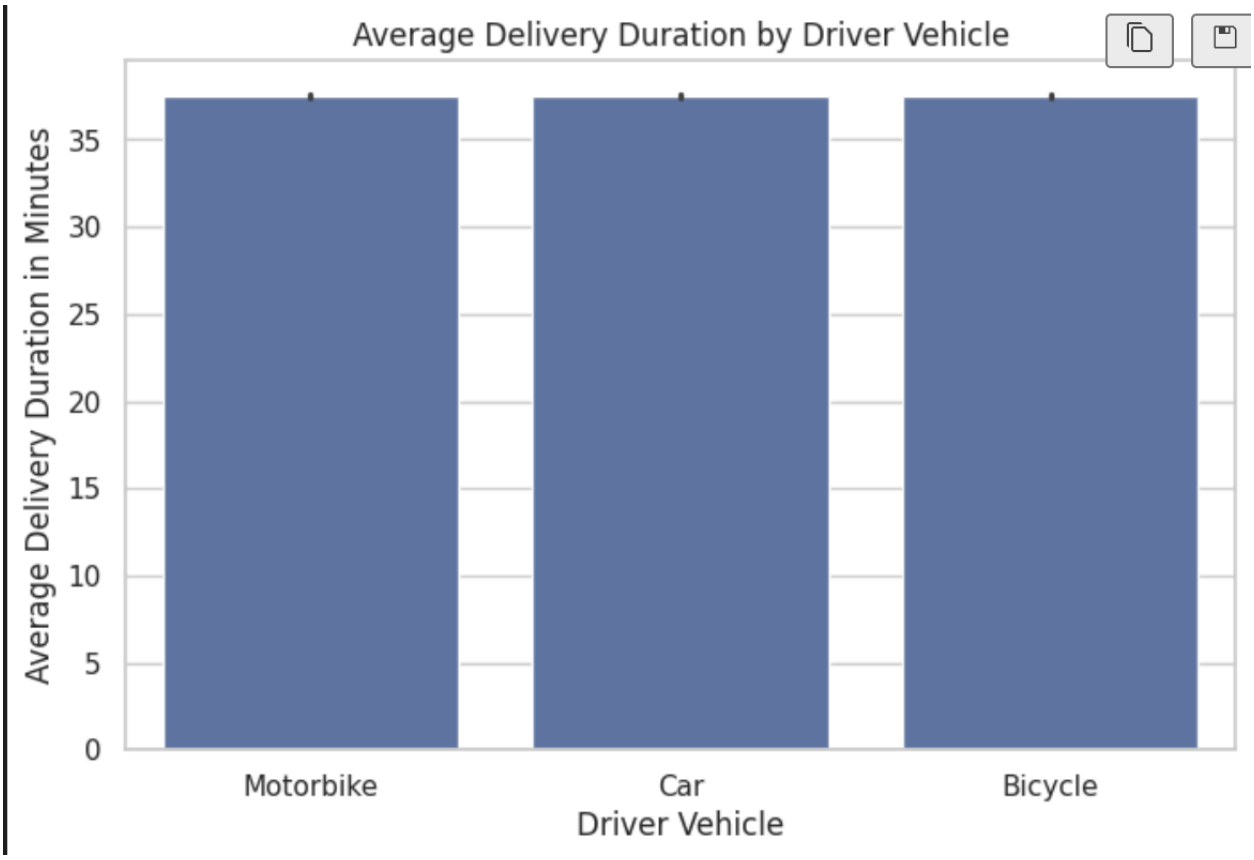
This graph shows the spread of delivery times in minutes for all orders. Delivery durations cluster around a central range (approximately 30-45 minutes), indicating consistent service times. There are few extreme delays, suggesting good operational efficiency.



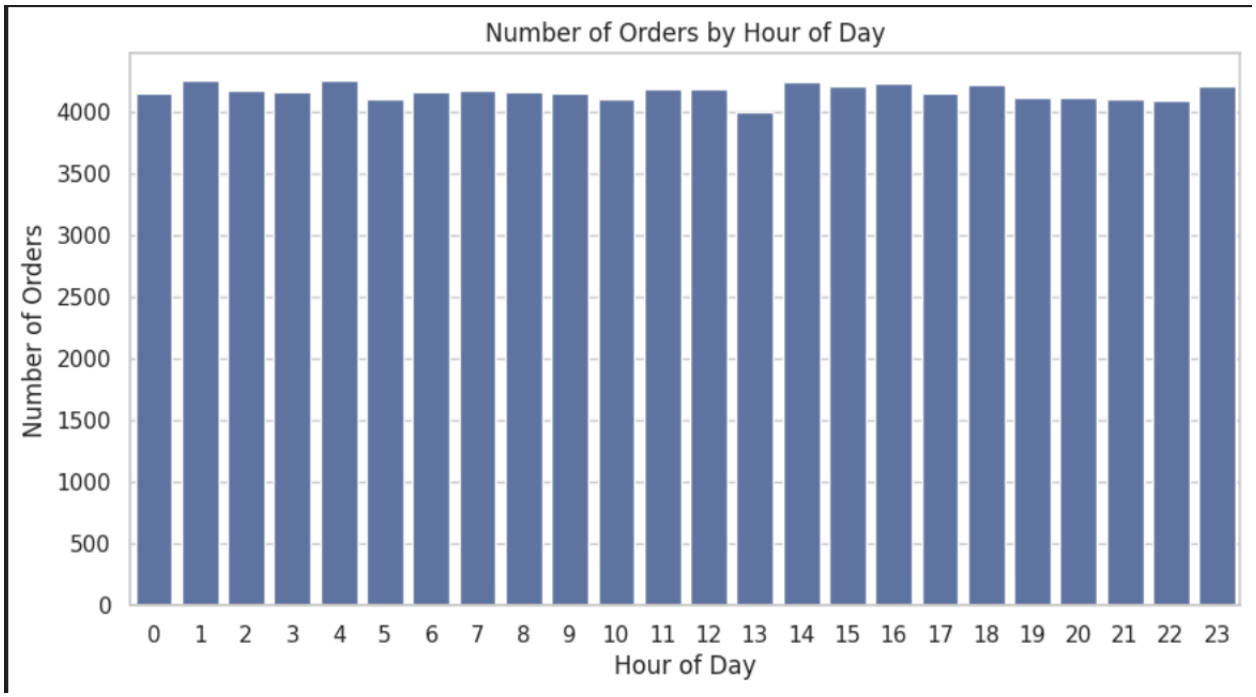
This boxplot compares how total price changes with the number of items ordered. There is a clear upward trend where total price increases as quantity increases. This confirms a strong and expected relationship between number of items and total cost.



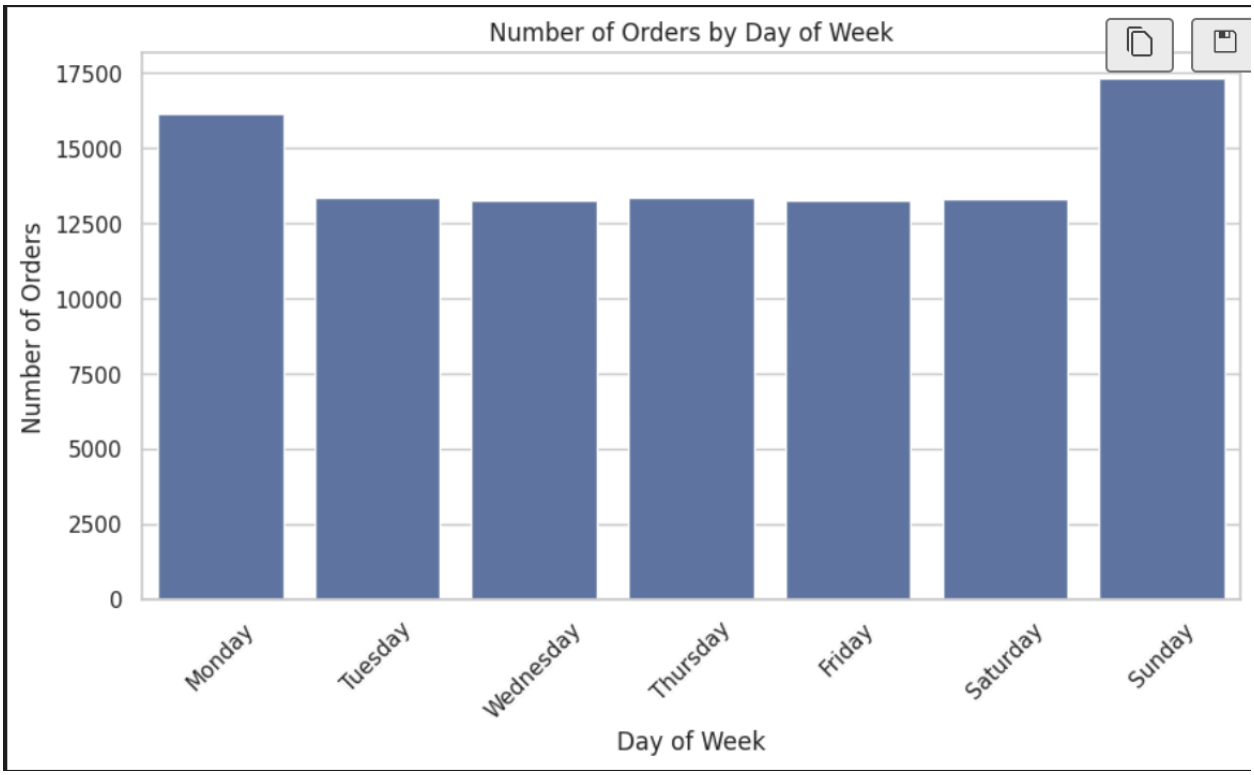
This bar chart shows the average delivery time under different traffic conditions. The average delivery duration is very similar across all traffic levels. This suggests that traffic does not significantly impact delivery time in this dataset.



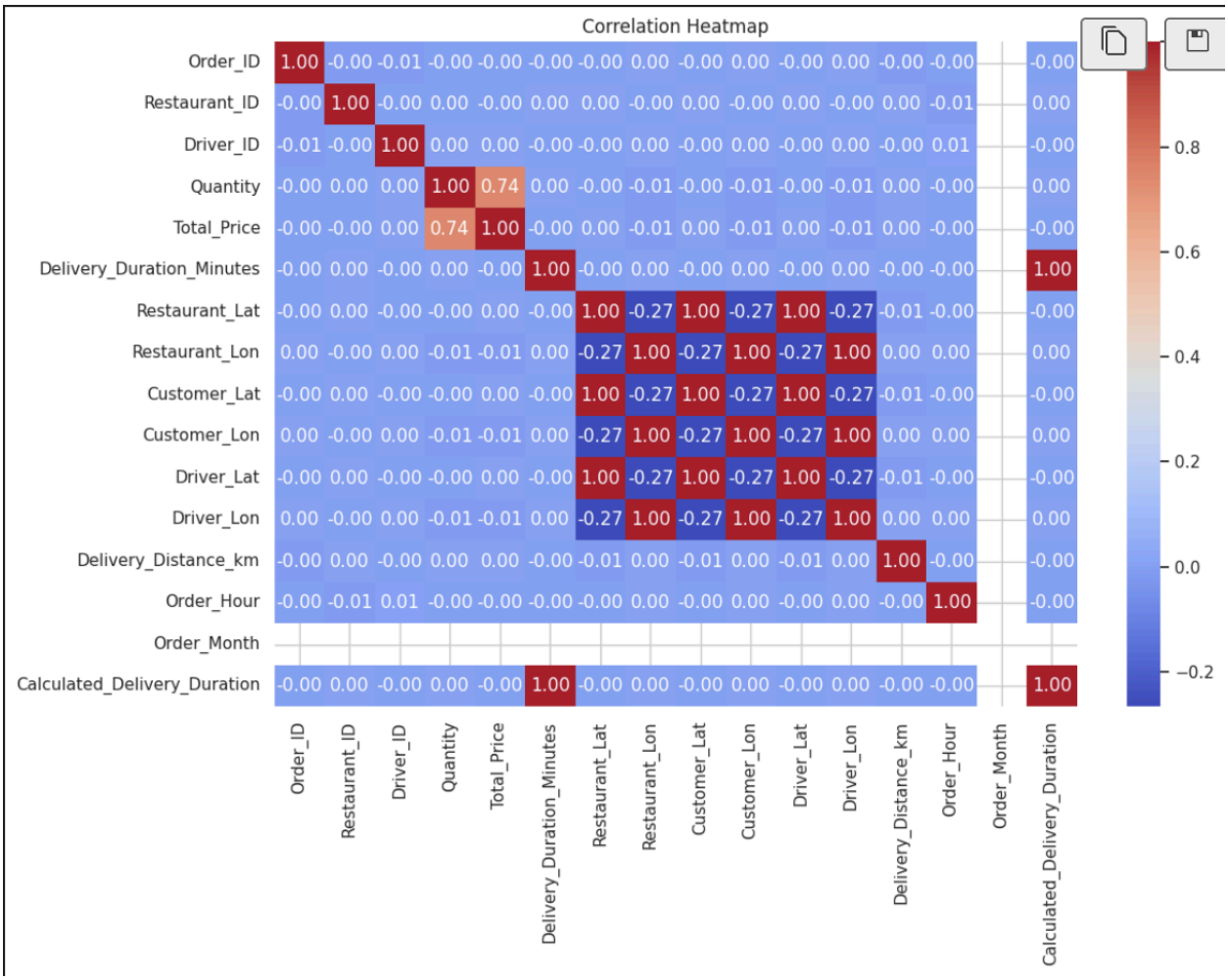
This graph compares delivery times based on the type of vehicle used (Car, Motorbike, Bicycle). Delivery durations are nearly identical across all vehicle types. This indicates that vehicle type does not play a significant role in delivery speed within this dataset.



This chart shows how order frequency changes throughout the day (0-23 hrs). Orders tend to increase during peak hours such as lunch and evening times. This reflects typical customer behavior where food demand is higher during meal times.



This visualization shows how orders are distributed across the days of the week. Orders are fairly evenly distributed across all days, with no strong peak on specific days. This suggests consistent demand throughout the week rather than weekend-heavy ordering behavior.



This heatmap shows the correlation between numerical variables such as price, quantity, distance, and delivery duration. Most variables show very weak correlations, indicating that they do not strongly influence each other. The only noticeable relationship between quantity and total price, which is expected.

Hypothesis Testing Steps

- Step 1: define hypothesis

  Example:

  - H0 (Null Hypothesis): No effect or no difference
  - H1 (Alternative Hypothesis): There is an effect
- Step 2: set significance level (α = 0.05)
- Step 3: choose test
  - t-test → comparing means

    - ANOVA → multiple groups
    - Chi-square → categorical relationships
    - Correlation → numerical relationships
- Step 4: compute p-value (using scipy library)
- Step 5: decision rule
    - If p-value < 0.05 → Reject H0
    - If p-value ≥ 0.05 → Fail to reject H0

Hypothesis testing results

Test 1: Delivery Duration vs 35 minutes

- p-value ≈ 0.000
- Decision: Reject H0

Conclusion: Delivery duration is significantly greater than 35 minutes.

Test 2: Traffic vs Delivery Duration

- p-value = 0.178
- Decision: Fail to reject H0

Conclusion: Traffic level does not significantly affect delivery duration.

Test 3: Payment Method vs Order Status

- p-value = 0.167
- Decision: Fail to reject H0

Conclusion: No relationship between payment method and order status.

Test 4: Distance vs Duration

- p-value = 0.596
- Decision: Fail to reject H0

Conclusion: No correlation between distance and delivery time.

Test 5: Quantity vs Total Price

- p-value ≈ 0.000
- Decision: Reject H0

Conclusion: Quantity significantly affects total price.

Conclusion

From this analysis:

- The dataset is clean and balanced
- Delivery times are consistent but slightly higher than expected
- Traffic does not significantly impact delivery time (unexpected finding)
- Distance also does not influence delivery time (possible data limitation)
- Quantity strongly affects total price (logical and expected result)

Any potential issues

Some limitations in the dataset:

1. **Synthetic or simulated data:** Relationships (e.g., traffic vs time) are unrealistic
2. **Lack of strong correlations:** Distance and duration should normally be related
3. **No missing data:** Real-world datasets usually contain missing values
4. **Uniform distributions:** Categories are too evenly distributed